



Fullstack Web Development

Session 1 | Week 1/Day 2 | 120 min

HTML5 - Part02

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HTML Forms

What is an HTML Form?

- An **HTML form** allows users to **enter data** on a webpage, which can be sent to a server for processing.
- It is created using the `<form>` element:

```
<form>  
  .  
  form elements  
  .  
</form>
```

HTML Forms

How the <form> Element Works

- Container for input elements such as:
 - Text fields
 - Password fields
 - Checkboxes
 - Radio buttons
 - Buttons (Submit / Reset)
- Wraps all the input elements together

```
<form action="submit.php" method="POST">  
  <label for="name">Name:</label>  
  <input type="text" id="name" name="user_name">  
  
  <label for="email">Email:</label>  
  <input type="email" id="email" name="user_email">  
  
  <input type="submit" value="Submit">  
</form>
```

HTML Forms

Important Attributes of <form>

Attribute	Description
action	URL where the form data is sent for processing
method	How the data is sent: GET or POST

```
<form action="submit.php" method="POST">
  <label for="name">Name:</label>
  <input type="text" id="name" name="user_name">

  <label for="email">Email:</label>
  <input type="email" id="email" name="user_email">

  <input type="submit" value="Submit">
</form>
```

HTML Forms

The action Attribute in HTML Forms

What It Does?

The **action attribute** specifies **where the form data should be sent** when the user clicks the **Submit** button.

- **Example:** Send the data to a server file called **submit.php**
- If **action** is **omitted**, the data is sent to the **current page** by default

HTML Forms

The action Attribute in HTML Forms

- Example Form:

```
<form action="/action_page.php" method="POST">
  <label for="fname">First Name:</label><br>
  <input type="text" id="fname" name="fname" value="John"><br>

  <label for="lname">Last Name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe"><br><br>

  <input type="submit" value="Submit">
</form>
```

When the user clicks Submit, the data (fname=John&lname=Doe) is sent to /action_page.php.

The server file can then process this data (save to database, send email,)

HTML Forms

The method **Attribute** in HTML Forms

What It Does?

The **method attribute** defines **how the form data is sent to the server**.

- **Two Common Methods**

Method	How It Works	Use Case
GET	Data is sent as part of the URL (query string: name=value&name2=value2)	Searching, simple data retrieval, bookmarking URLs
POST	Data is sent in the HTTP request body (not visible in URL)	Submitting sensitive information, large forms, login credentials

HTML Forms

The method Attribute in HTML Forms

- **Example** GET Method:

```
<form action="/submit.php" method="GET">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name">
  <input type="submit" value="Submit">
</form>
```

*If user enters "John", the URL becomes:
/submit.php?name=John*

Data is visible in the URL

**Default method is GET if you don't specify it*

HTML Forms

The method Attribute in HTML Forms

- **Example** POST Method:

```
<form action="/submit.php" method="POST">  
  <label for="name">Name:</label>  
  <input type="text" id="name" name="name">  
  <input type="submit" value="Submit">  
</form>
```

Data is sent in the HTTP request body

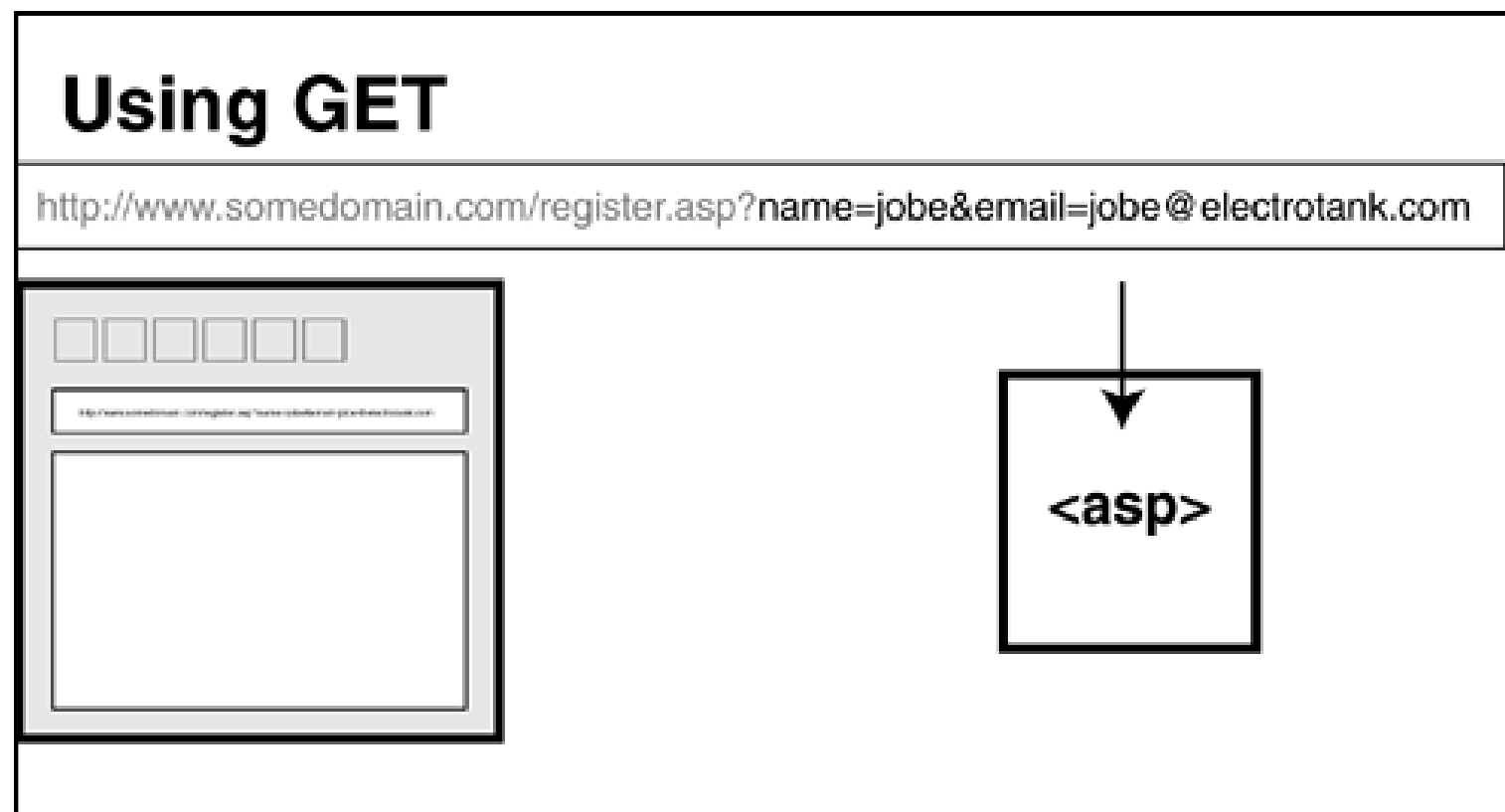
Not visible in the URL

More secure for sensitive information

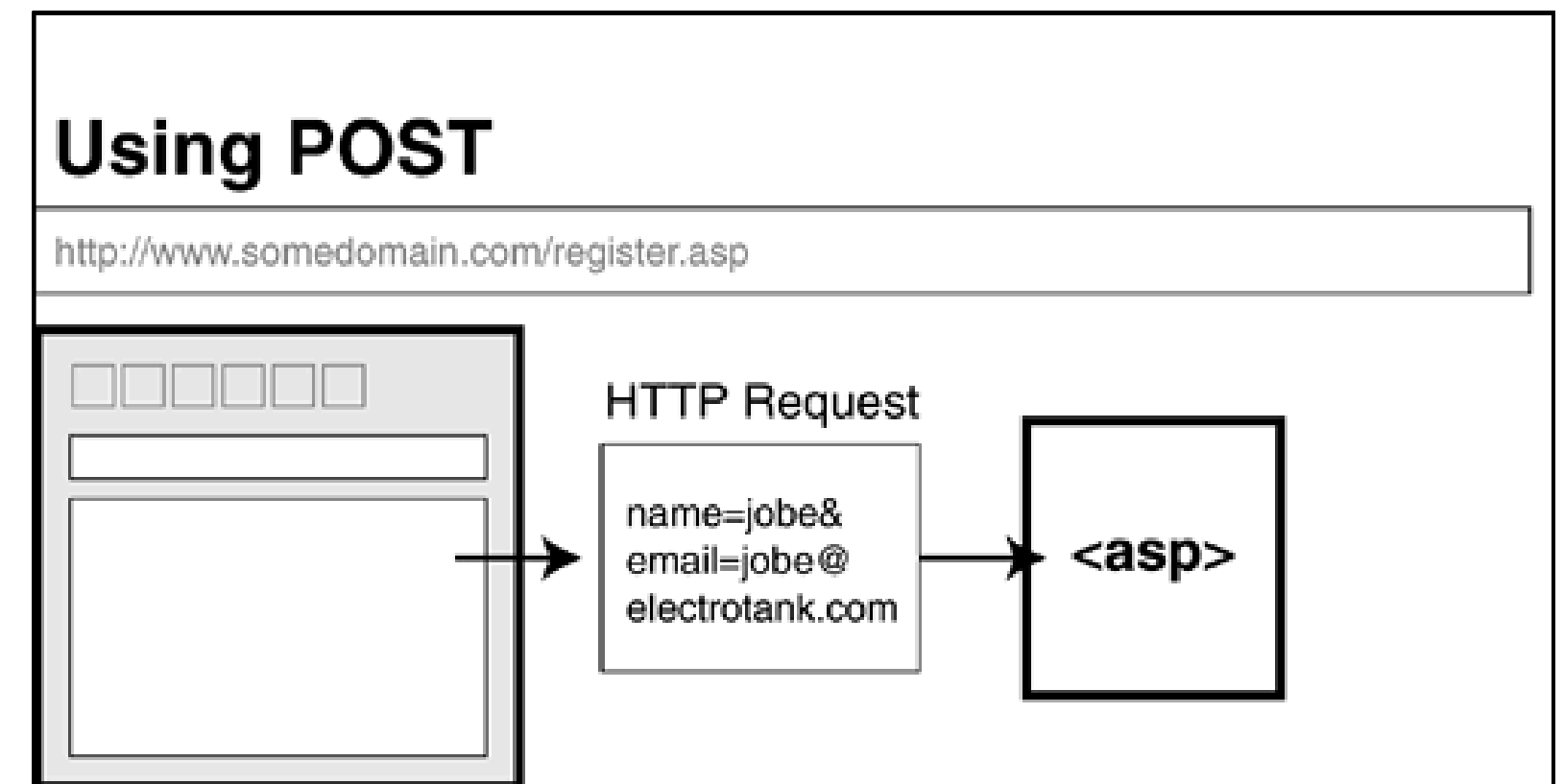
HTML Forms

The method **Attribute** in HTML Forms

- **Example :**



Data is visible in the URL



Not visible in the URL

HTML Forms

HTML Form Attributes : Submission Behavior

Attribute	Description	Example
action	URL where form data is sent	<code><form action="/submit.php"></code>
method	HTTP method used (GET or POST)	<code><form method="POST"></code>
enctype	How the data is encoded (only for POST)	<code><form enctype="multipart/form-data"></code>
accept-charset	Character encoding for form submission	<code><form accept-charset="UTF-8"></code>

HTML Forms

HTML Form Attributes : User Experience

Attribute	Description	Example
autocomplete	Enables/disables automatic field filling	<form autocomplete="off">
novalidate	Skips HTML validation on submission	<form novalidate>

HTML Forms

HTML Form Attributes : Form Identification & Linking

Attribute	Description	Example
name	Name of the form (used in scripts)	<form name="myForm">
rel	Relationship between linked resource & current page	<form rel="noopener">
target	Where to display the response	<form target="_blank">

HTML Forms

HTML <form> Elements

The **<form>** element can contain **various input and control elements**. Here's a breakdown:

Element	Description	Example
<input>	Single-line input field (text, password, checkbox, ...)	<input type="text" name="username">
<label>	Defines a label for an input (improves accessibility)	<label for="username">Username: </label>
<select>	Drop-down list	<select><option>Option 1</option> </select>

HTML Forms

HTML <form> Elements

The **<form>** element can contain **various input and control elements**. Here’s a breakdown:

Element	Description	Example
<textarea>	Multi-line text input	<textarea name="message"> </textarea>
<button>	Clickable button (submit, reset, or custom)	<button type="submit">Send</button>
<fieldset>	Groups related form elements	<fieldset><legend>Personal Info</legend>...</fieldset>
<legend>	Caption for a <fieldset>	<legend>Personal Info</legend>

HTML Forms

HTML <form> Elements

The **<form>** element can contain **various input and control elements**. Here’s a breakdown:

Element	Description	Example
<datalist>	Predefined list of options for an <input>	<input list="colors"><datalist id="colors"><option value="Red"></datalist>
<output>	Displays calculation results	<output name="result"></output>
<option>	Defines an option in <select> or <datalist>	<option value="1">One</option>
<optgroup>	Groups related options in a drop-down list	<optgroup label="Fruits"><option>Apple</option></optgroup>

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
text	One-line text input	<code><input type="text" name="username"></code>
password	One-line input for sensitive text; characters are hidden	<code><input type="password" name="pwd"></code>
submit	Sends form data to the server	<code><input type="submit" value="Send"></code>
reset	Resets all form fields to default values	<code><input type="reset" value="Clear"></code>
radio	Select one option from multiple choices	<code><input type="radio" name="gender" value="male"></code>

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
checkbox	Select multiple options	<code><input type="checkbox" name="hobby" value="reading"></code>
button	Simple push button (can be programmed with JavaScript)	<code><input type="button" value="Click me"></code>
file	Upload a file from user's device	<code><input type="file" name="resume"></code>
image	Graphical submit button (clickable image)	<code><input type="image" src="submit.png" alt="Submit"></code>
color	Input field for selecting a color	<code><input type="color" name="favcolor"></code>

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
date	Input field to select a date	<code><input type="date" name="birthday"></code>
datetime-local	Input field for date and time (no time zone)	<code><input type="datetime-local" name="meeting"></code>
email	Input field for email addresses	<code><input type="email" name="user_email"></code>
month	Input field to select month and year	<code><input type="month" name="expiry"></code>
number	Input field to enter a number	<code><input type="number" name="quantity" min="1" max="10"></code>

HTML Input Types

The **<input>** element is used to **collect user information**. Its behavior depends on the **type** attribute.

Type	Description	Example
url	Input field to enter a website URL	<code><input type="url" name="website"></code>
week	Input field for week and year	<code><input type="week" name="weeknum"></code>
search	Single-line field for search queries	<code><input type="search" name="searchbox"></code>
tel	Input field for telephone numbers	<code><input type="tel" name="phone"></code>

HTML Input Types

“I want to create a form where users can select one favorite color or multiple hobbies. Which input type should I use?”

text

checkbox

radio

submit

HTML Input Types Validation

HTML provides several **attributes to validate or control input fields**:

Attribute	Description	Example
maxlength	Maximum number of characters allowed	<code><input type="text" name="username" maxlength="10"></code>
min	Minimum value allowed (for number/date inputs)	<code><input type="number" name="age" min="18"></code>
max	Maximum value allowed (for number/date inputs)	<code><input type="number" name="age" max="99"></code>
readonly	Field cannot be edited by the user	<code><input type="text" name="id" value="123" readonly></code>

HTML Input Types Validation

HTML provides several **attributes to validate or control input fields**:

Attribute	Description	Example
required	Field must be filled before form submission	<code><input type="email" name="email" required></code>
disabled	Field is unusable and unclickable	<code><input type="text" name="promo" disabled></code>

HTML Layout (Semantic Elements)

HTML5 introduced **semantic layout elements** to describe the structure of a webpage clearly.

These elements help:

- Organize content
- Improve readability
- Improve SEO
- Improve accessibility

HTML Layout (Semantic Elements)

Common Layout Elements

Element	Description	
<header>	Field must be filled before form submission	<header>
<nav>	Field is unusable and unclickable	<nav>
<section>	Groups related content into sections	<section> <article> <aside>
<article>	Self-contained content (blog post, news article)	
<aside>	Sidebar content (ads, related links, info box)	
<footer>	Bottom section (copyright, contact info)	
<details>	Expandable content block	
<summary>	Visible heading for <details>	

HTML Layout Techniques

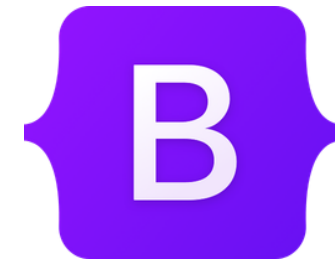
Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Frameworks

Pre-built CSS libraries that help build layouts quickly.

- Examples:
 - Bootstrap
 - Tailwind CSS
 - Foundation
- They provide:
 - Grid systems
 - Responsive design
 - Ready-made components



Foundation
Start here, build everywhere.

HTML Layout Techniques

Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Float (Older Technique)

- Originally used to wrap text around images
- Later used for page layouts
- Difficult to manage in complex layouts

Considered outdated for modern layouts.

```
.sidebar {  
    float: right;  
}
```

HTML Layout Techniques

Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Flexbox (Modern Layout)

- One-dimensional layout system
- Arranges items in a row or column
- Easy alignment and spacing
- Great for navbars, cards, small components

Best for Component-level layout.

```
.container {  
    display: flex;  
}
```


HTML Layout Techniques

Web page layout is controlled using CSS, not HTML alone.

There are **several techniques** to create layouts:

CSS Grid (Most Powerful)

- Two-dimensional layout system
- Controls rows and columns
- Perfect for full page layout

Best for Complex page structure.

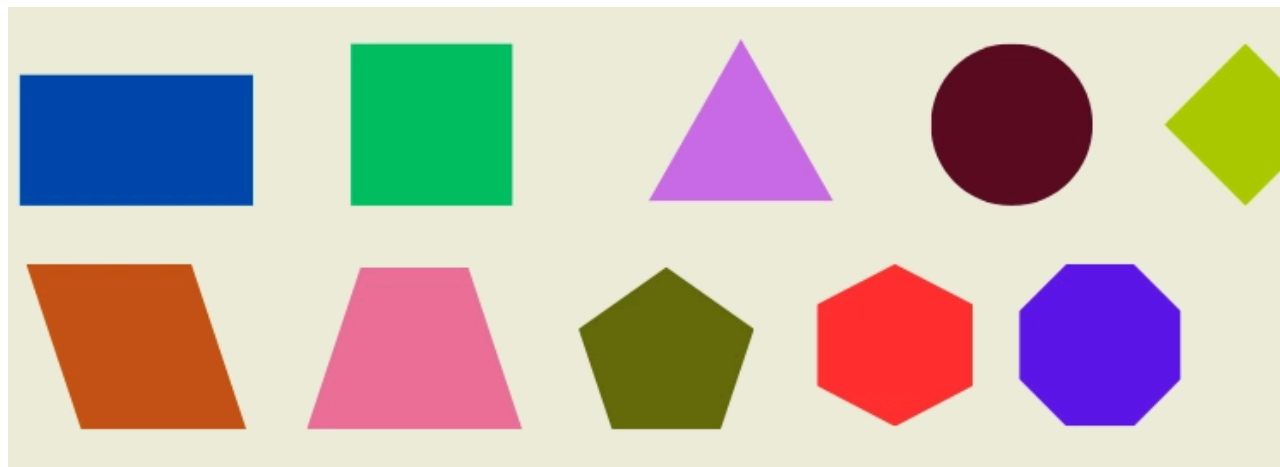
```
.container {  
    display: grid;  
    grid-template-columns: 1fr 1fr;  
}
```

HTML5 Graphics

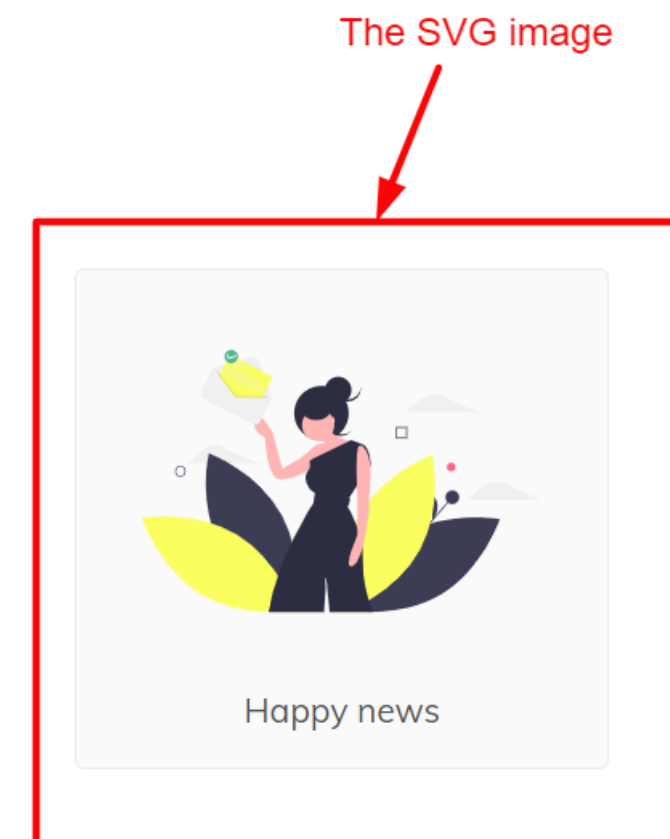
HTML5 provides **two main graphic** technologies:

- `<canvas>`
- `<svg>`

But they are used in different situations.



`<canvas>`



HTML5 Graphics – <canvas>

What is <canvas>?

The **<canvas>** element is used to **draw graphics dynamically** using JavaScript.

It can be used to create:

- Charts
- Games
- Animations
- Drawing applications
- Image manipulation

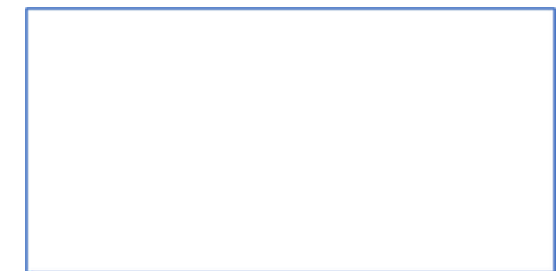
```
<canvas id="myCanvas" width="200" height="100"  
      style="border:1px solid #000000;">  
</canvas>
```

This only creates a blank rectangle.

Type : Pixel

<canvas> is just a blank drawing area.

Without JavaScript, it will only display an empty box.



HTML5 Graphics – SVG

What is SVG?

SVG (Scalable Vector Graphics) defines graphics using **XML-based markup**.

It is used to create:

- Logos
- Icons
- Charts
- Illustrations
- Scalable graphics

```
<svg width="100" height="100">  
  <circle cx="50" cy="50" r="40" stroke="green" stroke-width="4" fill="yellow" />  
  Sorry, your browser does not support inline SVG.  
</svg>
```

Type : Vector

- | | |
|--|--------------------------------|
| • <i><svg></i> → container | • <i>r</i> → radius |
| • <i><circle></i> → draws a circle | • <i>stroke</i> → border color |
| • <i>cx</i> → center X position | • <i>fill</i> → inside color |
| • <i>cy</i> → center Y position | |





THANK YOU

For your attention !

